UNIT 5 · ECONOMICS

Unit 5: Personal Inflation

Case-study · AI as a co-pilot

80 MINUTES · 4 ARTIFACTS

What we must have by the end

Data table

10 branded items · Jan 2024 vs Jan 2026 · 2 prices + 2 links each

Calculations

% growth per item · simple average · top-3 drivers

Rosstat comparison

Our basket estimate vs official CPI benchmark · explain the gap

2–3 final proposals

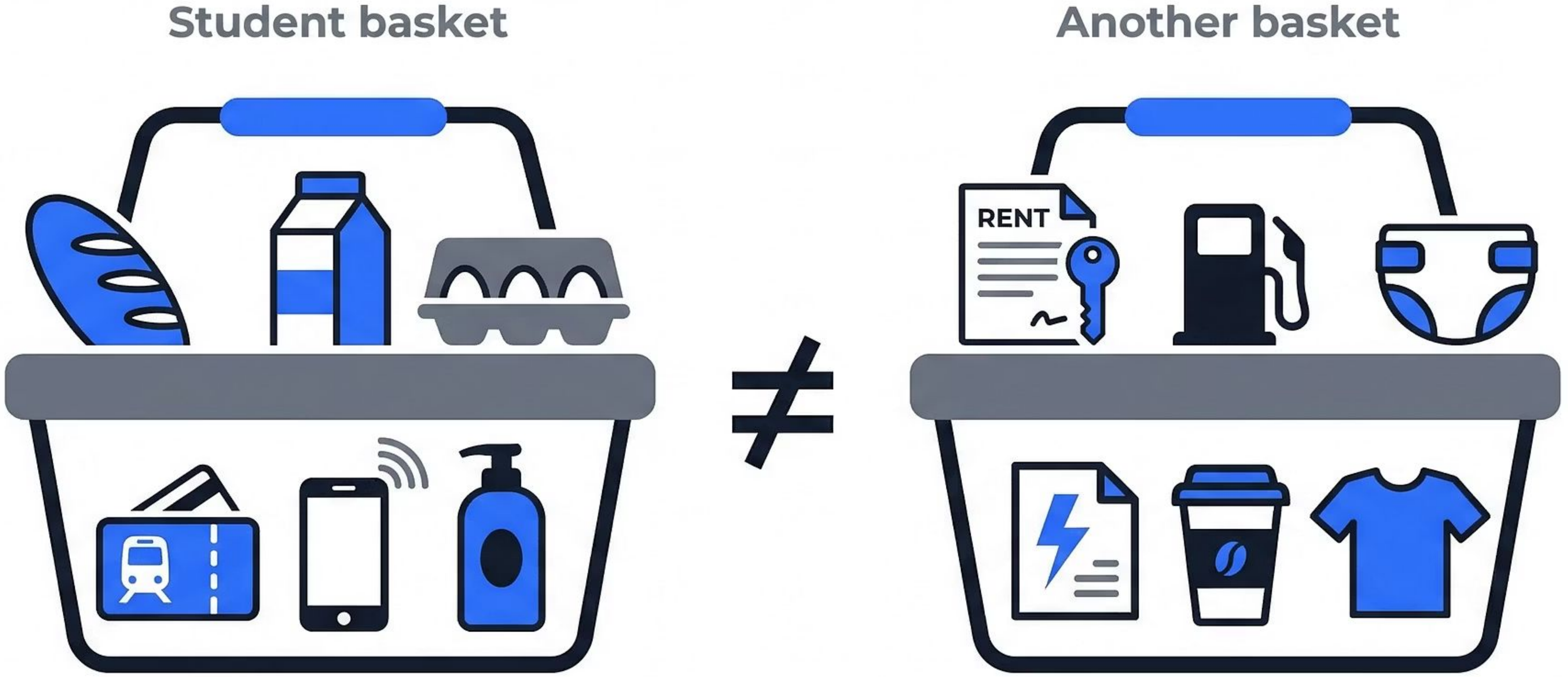
Each grounded in 1–2 specific drivers from our table

AI is a co-pilot for speed. You are responsible for validity and meaning.

Inflation: the simplest definition

- Inflation = prices go up (on average)
- Average always means: a basket
- So inflation is not one universal number

Key idea: "Same money buys less." The basket is the unit of measurement — change the basket, change the number.



Different baskets → different inflation

Official inflation vs personal inflation

Official CPI (Росстат)

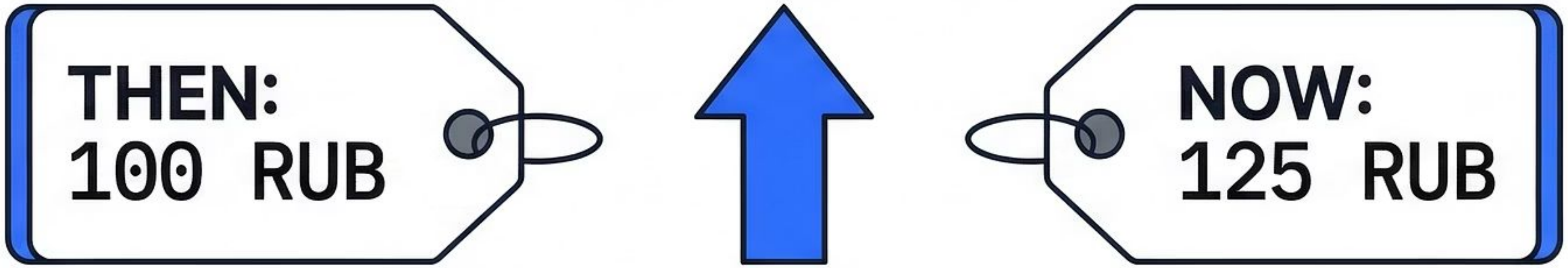
- Thousands of goods and services
- All of Russia
- Expenditure-weighted averages
- Published monthly
- Measures the average household

Personal basket estimate (this case)

- 10 branded items
- Moscow only
- Equal weights (teaching simplification)
- Jan 2024 → Jan 2026
- Measures a specific student profile

"Neither is 'wrong'. They measure different things. The gap between them is the analytical question."

How to compute price growth (%)



$$\text{Price growth, \%} = \left(\frac{\text{Price}_{\text{now}}}{\text{Price}_{\text{then}}} - 1 \right) \times 100\%$$

1

STEP 1:
Find relative change

Divide current price by the previous one.

Example: `125 / 100 = 1.25`

2

STEP 2:
Subtract one

Deduct 1 to isolate the growth factor.

Example: `1.25 - 1 = 0.25`

3

STEP 3:
Express as a percent

Multiply by 100 to get the percentage.

Example: `0.25 × 100% = 25%`

Basket inflation (our simple method)

01

Compute % growth for each of 10 items

Formula: $(\text{Price}_{2026} / \text{Price}_{2024} - 1) \times 100\%$

02

Take a simple average across all 10 items

Equal weights — every item counts the same

03

Identify the top-3 drivers

Items with the highest % growth

04

Interpret the result

What drove the basket? What does it mean for a Moscow household?

⚠️ **Teaching simplification:** equal weights is NOT the official CPI methodology. Rosstat uses expenditure-based weights. We use equal weights to keep the method transparent and computable in class.

This is NOT official CPI. This is a personal basket estimate — a teaching tool.

Open the case

Personal Inflation · Moscow · January 2024 → January 2026

What you have

- Shared Notion table (10 items, item names pre-filled)
- Browser + AI chat open
- 80 minutes

What you must produce

- Completed data table (10 items, 2 prices + 2 links each)
- Calculations (% growth, average, top-3)
- Rosstat comparison
- 2–3 final proposals

Read the case text on the next slide before starting. The case defines the question, the basket, and the constraints.

Case text

VERBATIM

> You are an analytics team hired by a Moscow NGO that supports students and young professionals. The organisation wants to understand how the real expenses of a typical Moscow student have changed over the past two years, and to prepare evidence-based support proposals.
>
> You are given a fixed basket of 10 goods and services regularly purchased by a Moscow student. Your task is to collect real prices for these items for January 2024 and January 2026, calculate personal basket inflation, compare it with the official Rosstat figure, and propose 2–3 concrete support measures backed by data.

Main question & basket inputs

VERBATIM

Main case question

> By what percentage did the real expenses of a Moscow student increase for this basket over the period January 2024 — January 2026? What explains the gap between your estimate and the official Rosstat inflation figure? What 2–3 support measures do you propose, and why?

Basket (10 items)

№	Item / service (EN)	Товар / услуга (RU)
1	Cheremushki sliced loaf 400 g	Батон нарезной «Черёмушки» 400 г
2	Domik v Derevne milk 2.5% 1 L	Молоко «Домик в деревне» 2.5% 1 л
3	Okskoye eggs C1 10 pcs	Яйца «Окское» С1 10 шт
4	Moscow Metro single ride Troika card	Метро Москва разовая поездка карта «Тройка»
5	MTS Tarifishche standard plan	МТС тариф «Тарифище» стандартный
6	ErichKrause A5 notebook 48 sheets grid	Тетрадь ErichKrause A5 48 листов клетка
7	Yandex Plus individual subscription 1 month	Яндекс Плюс индивидуальная подписка 1 месяц
8	Shokoladnitsa latte 300 ml	Латте 300 мл «Шоколадница»
9	O'STIN basic T-shirt no print	Футболка базовая O'STIN без принта
10	Head & Shoulders Classic Clean shampoo 400 ml	Шампунь Head & Shoulders Classic Clean 400 мл

All 10 items are fixed. Use the English name for source search in international retailers/archives; use the Russian name for Russian retailer websites.

Definition of Done

DOD · VERBATIM

- Data table: 10 items, Jan 2024 vs Jan 2026, 2 prices + 2 source links per item
- Calculations: % growth for each item + basket inflation as a simple average
- Interpretation: top-3 growth drivers + short explanation
- Benchmark: comparison with official Rosstat inflation for the same period
- Final position: 2–3 support measures linked to the basket results
- Optional: 1 visual (chart) if time allows

Live demo: 1 item → 2 prices + 2 links

01

Pick one item from the basket

State the exact spec: brand, product name, unit (e.g. 400 ml, 1 kg, 1 month subscription)

02

Find the Jan 2024 price from a specific retailer

Use the retailer's own website, archived page, or price aggregator. NOT Rosstat — Rosstat does not publish prices for branded items.

03

Find the Jan 2026 price from the same source

Same retailer preferred. If unavailable: comparable retailer in the same price tier. Same unit, same spec. No discounts / promo / loyalty prices.

04

Paste both prices + both links into the table

Price_{2024} · Price_{2026} · Source link 1 · Source link 2

Source fallback order: (1) Same archived source → (2) Official brand/retailer page → (3) Comparable retailer in the same price tier → (4) Price aggregator / archive as last resort. Document which fallback you used.

Why not Rosstat for item prices? Rosstat publishes category-level CPI averages across thousands of goods. It does NOT publish prices for Head & Shoulders, O'STIN, Яндекс Плюс, or any other branded item. Use retailers.

Case constraints

FIXED FOR ALL GROUPS · NON-NEGOTIABLE

Fixed basket

10 branded items — defined by the instructor. No substitutions.

Geography

Moscow, Russian Federation only

Periods

January 2024 vs January 2026. No other months.

Same spec

Same product: brand · size/unit · grade. Both periods must match exactly.

2 prices + 2 links

Per item. Both prices must have a source link.

No promo prices

No discounts · no sale prices · no loyalty card prices

If a constraint is unclear for a specific item, raise your hand immediately. Do not guess.

Data quality checklist

RUN THIS BEFORE SUBMITTING EACH ROW

✅ Same item?

Brand, product name, and unit match exactly in both periods

✅ Same unit?

Per kg, per litre, per piece, per month — identical in both rows

✅ Same retailer logic?

Same chain, or comparable chain in the same price tier

✅ Same month?

Both prices are from January (2024 and 2026 respectively)

✅ No discount?

Regular shelf price only — no promo, no sale, no loyalty

✅ Link saved?

Both source URLs are pasted into the table

If any answer is NO — fix the row before moving on. Do not proceed with bad data.

Class task (20 min): complete the table

- Each remaining item needs 2 prices (Jan 2024 & Jan 2026) + 2 source links
- Prices come from specific retailers or official brand/service pages — NOT Rosstat category averages
- Same retailer for both periods is preferred
- If the same retailer is unavailable: use a comparable retailer in the same price tier (keep it consistent across time)
- Same product spec + same unit required (brand, size, grade — no substitutions)
- No discounts / no promo prices / no loyalty card prices
- Timer: 20 minutes. Items assigned by row or small group to avoid duplication.

Source fallback order: same archived source → official retailer page → comparable retailer same price tier → aggregator/archive as last resort.

Checkpoint

Target

We need **10 / 10** items covered

Stuck?

Raise your hand — we assign immediately

Final calculations (with AI)

% growth per item

Formula applied to each row

Basket inflation

Simple average across all 10 items

Top-3 drivers

Items with highest % growth

AI prompt tip: Paste the completed table and ask AI to apply the formula to each row, compute the average, and list the top-3 by growth. Always specify the formula explicitly.

Sanity check (1 minute)

🚨 Hundreds of %?

Almost always a unit or period mistake — recheck that row

✅ Plausible range

Most items: roughly 5–50% over 2 years in this context

> "If a result looks impossible — it probably is. Check the input, not just the formula."

Official inflation benchmark (Rosstat)

Rosstat = official CPI benchmark ONLY. Our item prices come from retailers because the basket contains branded goods. Rosstat does NOT publish prices for specific branded items (Head & Shoulders, O'STIN, Яндекс Плюс, etc.). We use Rosstat only to compare our basket estimate against the official inflation figure.

01

Find the official CPI for the comparable period

Jan 2024 → Jan 2026 (or closest available) · rosstat.gov.ru or Bank of Russia dashboard

02

Record the figure + link

Paste into the Notion case page next to our basket result

Benchmark ≠ 'the truth'. Both numbers are valid for different purposes. Rosstat measures thousands of goods across Russia with expenditure weights. We measured 10 branded Moscow items with equal weights. The gap between the two numbers is itself an analytical finding.

Why our estimate may differ

- Different basket
- Different weights / method (simple average)
- Branded spec vs statistical categories
- Source choices

Generate candidate support measures

Rule: Each proposal must reference 1–2 specific drivers or items from our table.

01

Paste the top-3 drivers into AI

Include the % growth figures

02

Ask AI to generate 5–7 candidate measures

For a household / student facing this basket inflation

03

Review and narrow with the class

Select the 2–3 most realistic and evidence-linked options

Final position (2–3 measures)

1	2	3
What to do State the concrete action or policy	**Why** Explain the reasoning or mechanism	**Based on** Name the 1–2 drivers/items from the table

Format rule: No proposal is accepted without all three lines completed.

Quick checks

What were the top-3 drivers?

Name each item and its % growth

Why can personal inflation differ from official inflation?

Name at least two reasons from our case

HOMEWORK

Group homework: your own basket

Work in groups of 4–5. Build your own basket of 10 fully specified items — not the same as the class case. Follow the same pipeline: data → calculations → Rosstat comparison → 2–3 proposals.

01

Define 10 fully specified items

Brand · product name · unit · grade. Example: not 'milk' but 'Простоквашино whole milk 3.2% fat 1 litre'

02

Collect 2 prices + 2 source links per item

Jan 2024 vs Jan 2026 · Moscow · same retailer logic · no discounts

03

Calculate, compare, propose

% growth per item · simple average · Rosstat CPI benchmark comparison · 2–3 evidence-linked proposals

Deliverable: completed data table + calculations + Rosstat comparison + 2–3 proposals. Same format as the class case. Deadline: [to be announced].